

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) — Semester VI Examination, 2013-14

Paper No. MTB-605: Operations Research

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1.

- (a) Describe degeneracy in the solution of a transportation problem. [2]
- (b) Define a hyperplane and a polytope. [2]
- (c) Test whether $S = \{(x, y) : x \leq 4, y \geq 2\}$ is convex or not. [2]
- (d) Discuss loops and their properties in the context of solving a transportation problem. [2]
- (e) In the two-phase method for solving the LPP: Minimize $2x_1 + 3x_2$ subject to $4x_1 + 2x_2 \geq 12$, $x_1 + 4x_2 \geq 6$, $x_1, x_2 \geq 0$, construct the table for Phase-I. [2]
- (f) Prove that a hyperplane is a convex set. [2]
- (g) Consider optimization with equality constraints. Construct the Lagrange function and write the necessary conditions for stationary points. [2]
- (h) Write conditions on the Hessian matrix for a stationary point to be a saddle point. [2]
- (i) Write the dual of the primal: Maximize $Z = 3x_1 + 5x_2$ subject to $x_1 + 2x_2 + x_3 = 5$, $-x_1 + 3x_2 + x_4 = 2$, $x_1, x_2, x_3, x_4 \geq 0$. [2]
- (j) Transform into standard form: Minimize $Z = 2x_1 + 3x_2 - x_3$ subject to $|x_1 + x_2| \leq 4$, $x_1 + x_2 + x_3 \geq 5$, $x_1 \geq 0$, x_2, x_3 unrestricted. [2]
- (k) Examine whether $(1, 0, 1)$ is a basic solution of $x_1 + x_2 + x_3 = 2$, $x_1 - x_2 + x_3 = 2$. [2]

Question 2.

- (a) Prove that the set of all convex combinations of a finite number of points of a set $S \subset \mathbb{R}^n$ is a convex set. [5]
- (b) Let $x_1 = 1$, $x_2 = 2$, $x_3 = 1$, $x_4 = 0$ be a feasible solution to $11x_1 + 2x_2 - 9x_3 + 4x_4 = 6$, $15x_1 + 3x_2 - 12x_3 + 5x_4 = 9$. Reduce this to a basic feasible solution. [7]

Question 3.

- (a) Show that the set of all feasible solutions to the LPP $Z = C^T x$ subject to $Ax = b, x \geq 0$ is a convex set. [5]
- (b) Solve the LPP using the Big-M (penalty) method: Maximize $Z = 2x_1 + x_2 + 3x_3$ subject to $x_1 + x_2 + 2x_3 \leq 5, 2x_1 + 3x_2 + 4x_3 = 12, x_1, x_2, x_3 \geq 0$. [7]
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Question 4.

- (a) Prove that the dual of a dual is the primal. [5]
- (b) Find the dual of the LPP: Maximize $Z = 5x_1 + 12x_2 + 4x_3$ subject to $x_1 + 2x_2 + x_3 \leq 5, 2x_1 - x_2 + 3x_3 \leq 2, x_1, x_2, x_3 \geq 0$, and hence solve it. [7]
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Question 5.

- (a) Find the Initial Basic Feasible Solution (IBFS) by Vogel's Approximation Method (VAM) and the matrix minima method for the transportation problem below. Do they differ?

	D_1	D_2	D_3	D_4	D_5
Supply					
S_1 8	2	3	4	5	6
S_2 9	3	4	5	6	7
S_3 13	4	5	6	7	8
Demand	10	3	4	13	

[5]

- (b) Solve the LPP using the revised simplex method: Maximize $f = -4x_1 - 3x_2$ subject to $5x_1 + x_2 \geq 11, -2x_1 - x_2 \leq -8, x_1 + 2x_2 \geq 7, x_1, x_2 \geq 0$. [7]
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Question 6.

- (a) State and prove the necessary condition(s) for extreme points of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ with continuous partial derivatives. [5]
- (b) Using Karush-Kuhn-Tucker (KKT) conditions, solve: Maximize $f(\mathbf{x}) = x_1^2 + x_2^2 + 5x_1x_2x_3$ subject to $x_1^2 - x_2^2 + x_3^2 = 10, x_1^2 - x_2^2 + 4x_3^2 \geq 20$. [7]
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Question 7.

- (a) Solve the assignment problem to *maximize* the expected return of assigning a job to a machine:
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	M_1	M_2	M_3
M_4			
J_1	42	35	28
21			
J_2	30	25	20
15			
J_3	30	25	20
15			

[5]

- (b) Find the least-cost transportation schedule using the NWC method for IBFS and then the UV (MODI) method for the optimal allocation:

	W_1	W_2	W_3	W_4
Supply				
F_1	45	24	21	12
500				
F_2	35	20	14	16
400				
F_3	42	16	21	19
710				
Demand	300	250	350	400

[7]